AI MVP Decision Log and Revised Module Architecture

This document records the AI module path taken so far and proposes a revised MVP architecture for the generation and analysis modes.

1. Logging: Past Work, Roadmap, Decisions, Results, and Analysis

Project Target

The project is a research-oriented prototype for speculative ecoacoustic soundscape generation. The core goal is not to reproduce one exact real recording from environmental data, but to generate plausible soundscapes under altered environmental conditions.

Current MVP focus:

- Prioritise the AI module over frontend/backend polish.
- Use the existing React/Vite frontend and Express backend as integration shells.
- Build an AI path that can produce believable audio and explainable outputs for demo and research evaluation.

Current System State

Area	Current State	Notes
Frontend	Basic React/Vite UI exists	Generation and transformation pages are scaffolded.
Backend	Express API exists	Auth endpoints and AI proxy routes exist. Auth is temporarily disabled in `requireAuth`.
AI server	FastAPI server exists	`acoustic_ai/server.py` exposes `/health`, `/analysis`, and `/generation`.

Area	Current State	Notes
Dataset	Site 257 Bowra Dry-A MVP sample exists	287 recordings, 6,148 usable clips, environmental data aligned.
Model checkpoint	VAE checkpoint exists	`acoustic_ai/checkpoints/best.pt` .
Latent database	Per-clip latent DB exists	`acoustic_ai/latent_clips.npy` .
Vocoder	Ecoacoustic HiFi-GAN checkpoint exists	`acoustic_ai/vocoder_checkpoints/best.pt` .

Dataset and Preprocessing Roadmap

Step	Decision	Result	Analysis
Use Site 257 Bowra Dry-A	Start with one known ecoacoustic site instead of many sites	Reduced scope and improved consistency	Good MVP decision. A single site limits domain variation and makes evaluation easier.
Use diel stratified sampling	Sample across dawn, morning, afternoon, and night	287 recordings across 73 local dates	Preserves time-of-day variation, which is one of the strongest ecoacoustic drivers.
Use clips instead of full FLAC files	Download shorter webm	6,148 clips available	More practical for training.

Step	Decision	Result	Analysis
	clips for training		Original long files are not needed for MVP.
Align environmental data per recording	Fetch NASA POWER hourly/daily variables	`site_257_env_data.csv` aligned to 287 recordings	Useful conditioning data, but coarse and indirect for many acoustic effects.
Build training manifest	Join clip paths with environmental data	`site_257_training_manifest.csv` with 6,148 rows	Critical for reproducible training.
Convert/precompute audio	Convert webm to wav and precompute mel spectrograms	`.webm.wav` and `.npy` spectrogram paths used by dataset	Good for iteration speed, but naming is non-standard and should be documented.
Exclude unrecoverable clips	Do not retry 12 permanent A2O `422` failures	12 missing clips excluded	Negligible impact: 12 out of 6,160 candidate clips.

Environmental Feature Design

Current model input uses 29 environmental features:

- 15 numeric features such as temperature, humidity, wind speed, precipitation, solar radiation, pressure, daily min/max temperature, days since rain, daylight hours, and local/UTC hour.

- 3 circular features encoded as sin/cos: wind direction, month, day of year.
- 2 categorical features encoded as one-hot: season and sample bin.

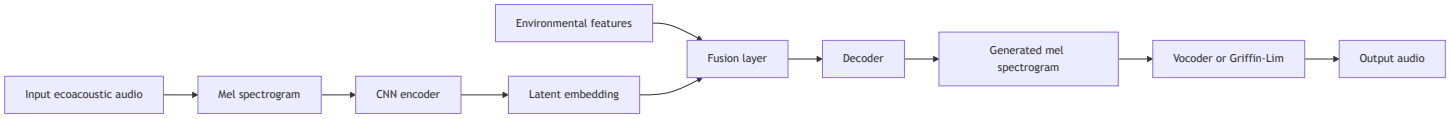
Important interpretation:

- Wind speed and rainfall are direct acoustic drivers because they can produce audible sound.
- Temperature, humidity, pressure, daylight, season, and days since rain are indirect ecological drivers.
- Indirect variables can influence species activity and vegetation state, but they do not map cleanly to a waveform by themselves.

Model Roadmap So Far

Attempt 1: Plain Autoencoder

Original idea:



Decision:

- Use an autoencoder to learn a latent representation of environmental soundscape structure.
- Combine audio embedding with environmental variables.
- Decode latent representation back into a mel spectrogram.

Result:

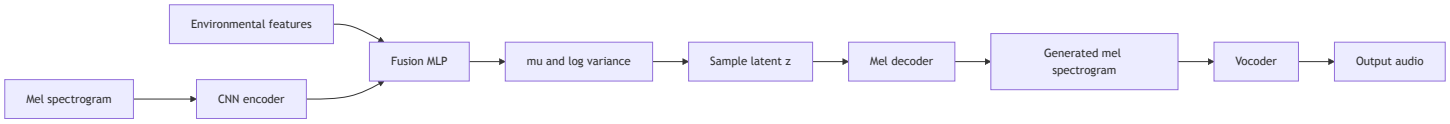
- Reconstruction/generation quality did not meet expectations.
- Generated audio sounded like electronic machine noise rather than plausible ecoacoustic audio.

Analysis:

- A plain autoencoder learns to reconstruct training inputs, but it does not guarantee that averaged or sampled latent points decode to meaningful spectrograms.
- Mean latent templates can land in regions the decoder has never seen during training.
- Ecoacoustic audio often contains noise-like textures, silence, wind, insects, distant calls, and recording artefacts. MSE reconstruction can average these into blurred spectrograms.

Attempt 2: VAE Module

Updated idea:



Decision:

- Switch from a plain autoencoder to a VAE.
- Add KL regularisation so the latent space is smoother and more sampleable.
- Use a decoder with a larger temporal seed to avoid flat time-axis output.

Implementation details:

- `acoustic_ai/model.py` defines `SoundscapeModel`.
- CNN encoder maps mel spectrograms to a 512-dim audio embedding.
- Fusion MLP combines the audio embedding with 29 env features.
- VAE heads produce `mu` and `log_var`.
- Latent dimension is 256.
- Decoder outputs 128-bin mel spectrogram crops.
- Training objective is `MSE(reconstruction, input) + beta * KL`, with beta documented as 0.01.

Result:

- Better model structure than the plain autoencoder.
- Still did not produce believable ecoacoustic sound when used as a direct generator.
- Audio remained machine-like or noisy.

Analysis:

- The VAE improved the latent space but did not solve the main identifiability problem.
- Coarse environmental data cannot uniquely define a full ecoacoustic waveform.
- The model is asked to learn too many things at once: background ambience, wind, rain, species calls, recording conditions, microphone artefacts, and temporal event structure.
- MSE-style spectrogram reconstruction still encourages averaged, blurry outputs.
- Random or weakly grounded latent sampling remains risky.

Attempt 3: Nearest-Neighbour Latent Grounding and Ecoacoustic Vocoder

Updated generation path:

Decision:

- Stop relying on pure latent sampling.
- Use the per-clip latent database `latent_clips.npy` .
- Retrieve the top-k training clips whose environmental vectors are most similar to the requested conditions.
- Average their latent vectors and add controlled noise.
- Use an ecoacoustic HiFi-GAN trained on Site 257 audio instead of a speech-domain vocoder.

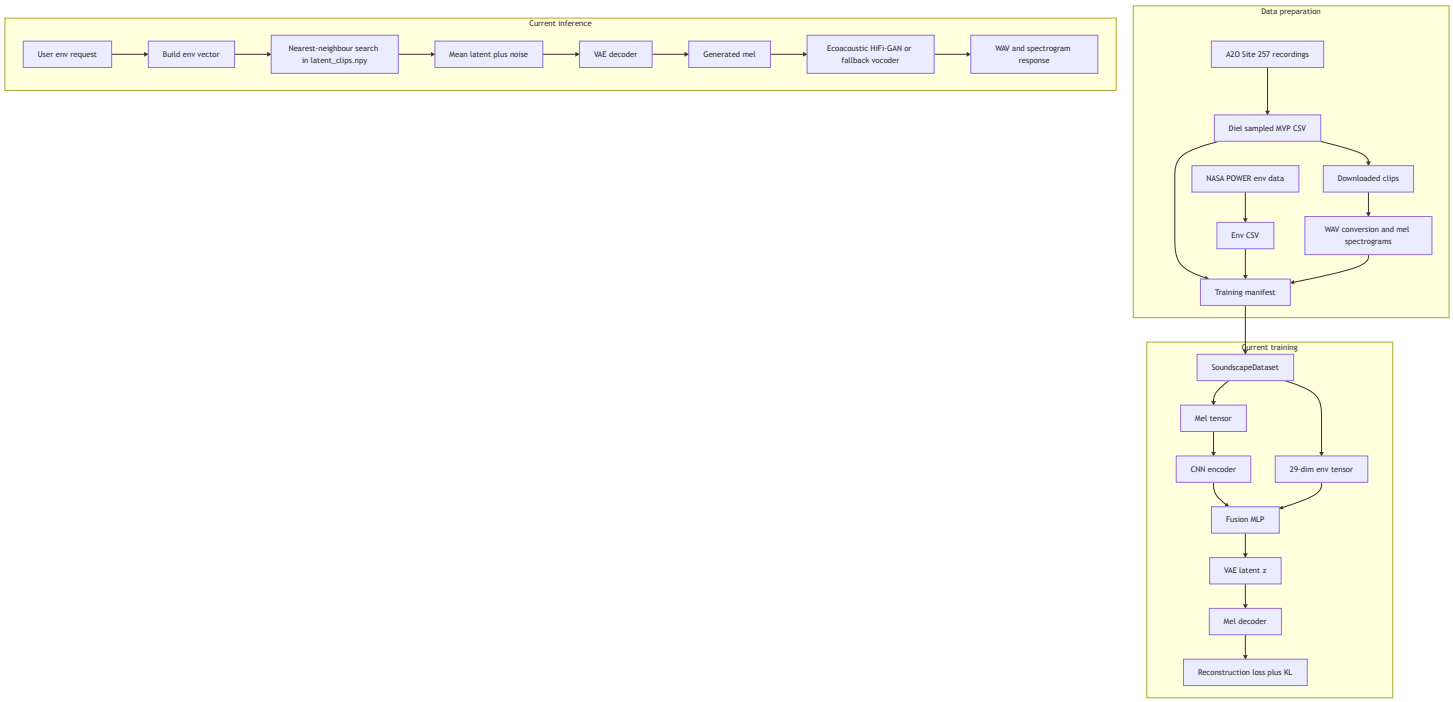
Result:

- More grounded than direct VAE sampling.
- The generated output is still limited by the decoder and latent representation.
- The vocoder path is more appropriate because it matches 128 mel bins and 22,050 Hz ecoacoustic audio.

Analysis:

- This is a better fallback strategy because generation starts near real training examples.
- However, it still tries to produce the whole soundscape through one latent vector.
- It does not explicitly separate background bed, weather sound, species events, and other annotated events.
- It has limited temporal/ecological narrative because a 30-second latent crop is static.

Current AI Architecture Picture



Main Lessons From Past Work

Lesson	Implication for MVP
Full soundscape generation from coarse env data is underdetermined	Do not rely on one model to generate all acoustic content directly from weather variables.
Temperature and humidity are mostly indirect acoustic variables	Use them for event/species likelihood, not direct waveform generation.
Wind and rainfall are directly audible	Model or mix them as separate controllable layers.
Species calls and annotations have clearer ecological links	Treat annotated events as a separate event layer.
VAE latent sampling is risky for audio quality	Prefer retrieval, latent grounding, or asset-based mixing for MVP.
Vocoder domain matters	Ecoacoustic vocoder is better than speech-domain HiFi-GAN for this project.
A single generated spectrogram hides too many factors	Layered generation gives better control, explanation, and evaluation.

2. New Approach: Generation Mode and Analysis Mode

New MVP Principle

The revised MVP should treat a soundscape as layered acoustic composition rather than one monolithic generated waveform.

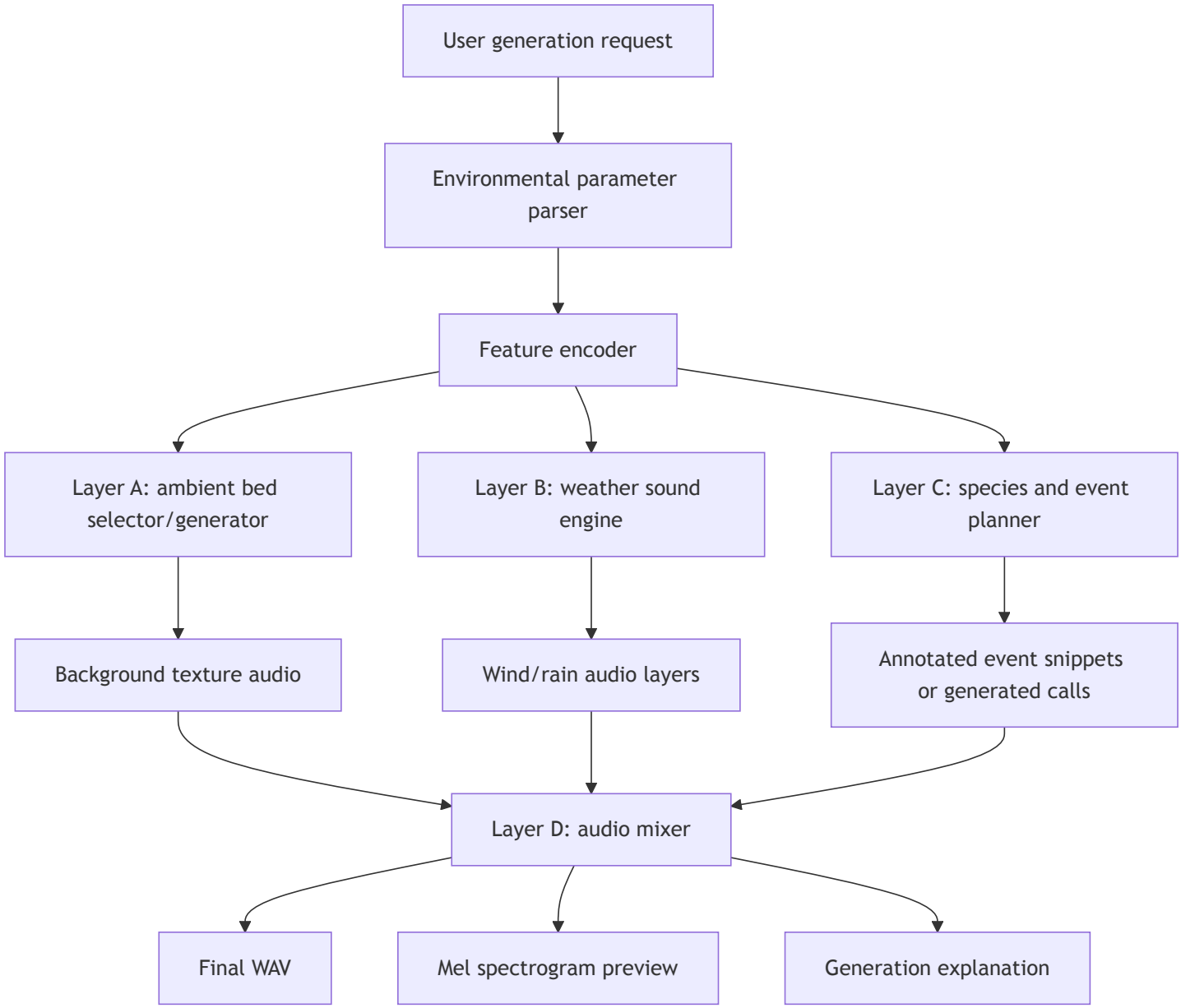
Proposed soundscape equation:

```
speculative soundscape
= ambient site bed
+ weather layer
+ biological/event layer
+ final mix and explanation
```

This better matches the real causal structure:

- Ambient bed captures general site texture.
- Wind and rain are directly controlled by weather values.
- Species and annotated events are controlled by season, time, temperature, humidity, rainfall history, and labels.
- The final mixer produces the user-facing audio.

Generation Mode: Revised Architecture



Generation Layer A: Ambient Site Bed

Purpose:

- Provide the continuous ecoacoustic background texture.
- Capture general site tone, insects, low-level ambience, microphone characteristics, distant environmental noise, and non-specific acoustic texture.

Recommended MVP implementation:

- Use retrieval first.
- Given user conditions, find similar real clips by season, sample bin, and environmental vector.
- Use the retrieved audio directly or use VAE reconstruction/transformation only as a secondary option.

Why:

- This gives realistic audio immediately.
- It avoids asking the VAE to invent a whole soundscape from weak conditioning.
- It remains research-valid because the bed is selected based on environmental similarity.

Comparison:

Approach	How it works	Pros	Cons	MVP fit
Raw audio retrieval	Pick one or more real clips similar to requested env conditions	Highest realism, fast, explainable	Less novel generation; limited to dataset examples	Strong
Latent nearest-neighbour plus VAE decode	Retrieve similar latent vectors, decode through VAE	Uses existing model, more generative	Still risks machine-noise output	Medium
VAE random sampling	Sample latent and decode	Simple once trained	Poor quality and weak control observed	Weak
Diffusion background generator	Train conditional diffusion for ambience	Best long-term generative direction	Too expensive and risky for MVP	Future

MVP recommendation:

```
Use raw/latent retrieval for ambient bed.
Keep VAE as representation and transformation support, not the only renderer.
```

Generation Layer B: Weather Sound Engine

Purpose:

- Generate or mix direct weather sounds: wind and rain.
- Map user parameters to audible intensity, texture, and volume.

Inputs:

- `wind_speed_ms`
- `wind_direction_deg`
- `wind_max_ms`
- `precipitation_mm`
- `precipitation_daily_mm`
- `days_since_rain`
- optional: `humidity_pct` , `sample_bin` , `season`

Recommended MVP implementation:

- Build a small library of real wind and rain clips from the dataset.
- Tag or bucket them by intensity.
- Mix them over the ambient bed with parameter-controlled gain, filtering, and density.

Comparison:

Approach	How it works	Pros	Cons	MVP fit
Rule-based asset mixing	Select wind/rain assets by intensity and mix them	Very controllable, good audio quality, easy to explain	Requires curated assets	Strong
Classifier plus retrieval	Detect wind/rain intensity in existing clips, retrieve matching layers	More data-driven	Needs labels or heuristics	Strong after curation
Procedural synthesis	Generate wind/rain from noise filters and envelopes	No dataset dependency, parameter-controllable	May sound artificial if not tuned	Medium

Approach	How it works	Pros	Cons	MVP fit
Neural weather generator	Train model for rain/wind conditioned on values	More advanced	Needs labelled/intensity-specific data	Future

MVP recommendation:

```
Use curated real weather assets plus simple parameter-controlled mixing.
```

Example mapping:

Parameter condition	Weather layer behaviour
`wind_speed_ms < 2`	No wind layer or very low gain.
`2 <= wind_speed_ms < 6`	Light wind bed, high-pass filtered less aggressively.
`wind_speed_ms >= 6`	Stronger wind layer, more low-mid energy and amplitude movement.
`precipitation_mm == 0`	No rain layer.
`0 < precipitation_mm < 2`	Sparse light rain layer.
`precipitation_mm >= 2`	Dense rain layer with higher gain and broader spectrum.

Generation Layer C: Species and Annotated Event Layer

Purpose:

- Add recognisable biological or annotated acoustic events.
- Make environmental variables meaningful through ecological likelihood, not direct waveform control.
- Treat annotation data as a preprocessing dependency: the final event-layer design cannot be fixed until the annotation files are audited and converted into a usable event index.

Pre-training requirement: annotation audit and preprocessing

Before training any species/event model, the project needs a dedicated annotation analysis step. The current A2O annotation CSVs contain mixed event types and mixed label quality, so the first task is to

identify what labels are actually available and how reliable they are.

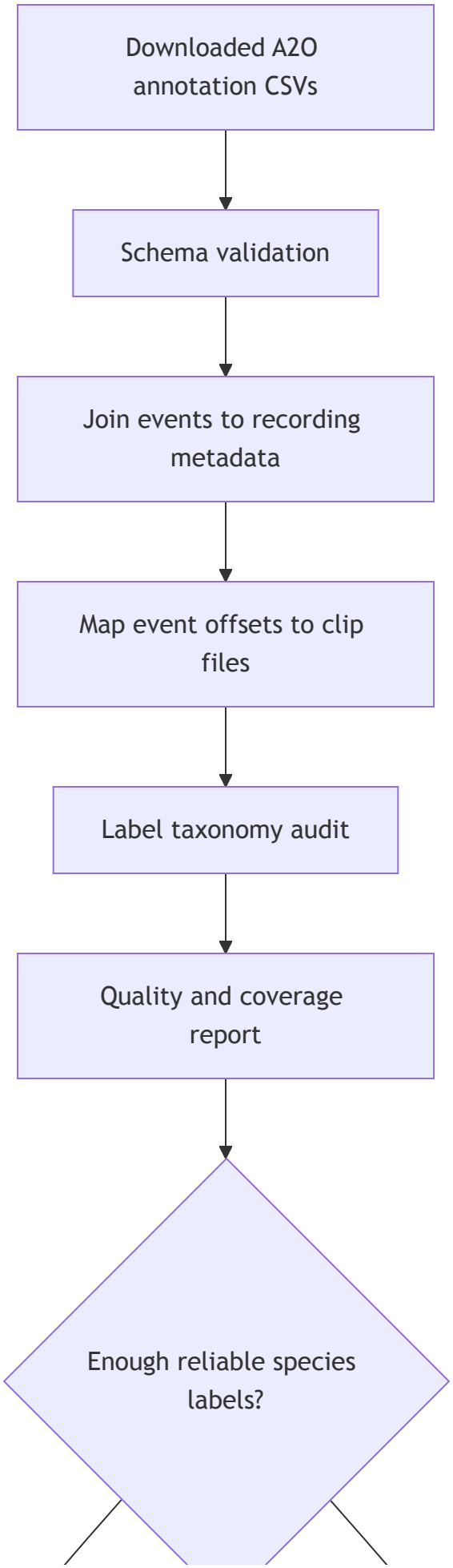
Current annotation schema includes:

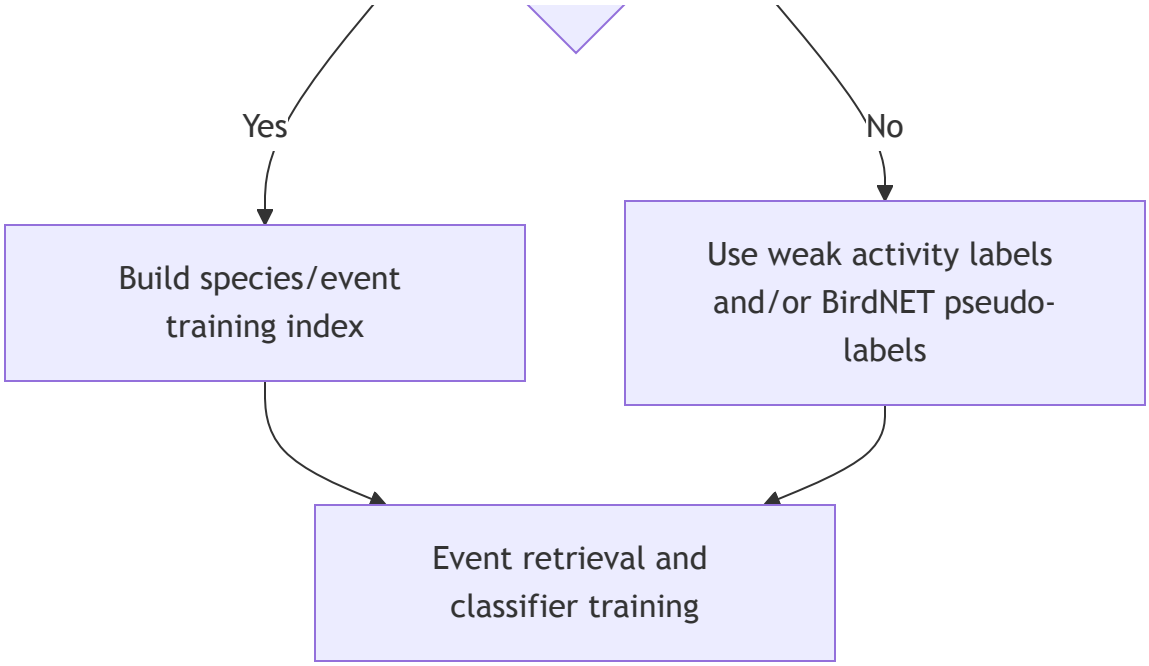
Field group	Example columns	Why it matters
Recording identity	`audio_recording_id` , recording start date/time, site fields	Joins each event back to clips, env data, and source audio.
Event timing	`event_start_seconds` , `event_end_seconds` , `event_duration_seconds`	Required to extract event snippets and align events to 300 s clips.
Frequency region	`low_frequency_hertz` , `high_frequency_hertz`	Useful for filtering, event-type grouping, and spectrogram cropping.
Confidence/activity	`score` , `is_reference`	Can support thresholding and weak activity labels.
Biological labels	`common_name_tags` , `species_name_tags`	Candidate supervised labels for species/event layer.
Other labels	`other_tags`	May contain non-species acoustic classes or useful metadata.
Verification	`verification_*` , `verification_consensus`	Determines whether labels should be trusted as ground truth or weak labels.
Import/source	`audio_event_import_file_name` , `audio_event_import_name`	Helps distinguish BirdNET imports, manual annotations, and other sources.

Known data issue from the current audit notes:

Observation	Impact
287 annotation files exist, but only a subset contains event rows	Annotation availability is sparse at recording level.
2,739 event rows were previously reported across 22 recordings	Species/event training data is much smaller than the audio dataset.
Many events are score-only with no common/species name	These are useful as weak `bioacoustic_activity` labels, but not species labels.
Species-labelled events are imbalanced	A classifier trained directly on raw labels may overfit dominant species/classes.
Some labels appear to come from BirdNET imports	Label source and confidence threshold must be tracked.

Annotation preprocessing workflow:


Downloaded A2O annotation CSVs
Schema validation
Join events to recording metadata
Map event offsets to clip files
Label taxonomy audit
Quality and coverage report
Enough reliable species labels?



Required preprocessing outputs:

Output	Description	Used by
`annotation_event_index.csv`	One row per usable event with recording ID, clip path, event start/end, label, score, source, verification status, and env features	Event retrieval, event scheduling, classifier training
`annotation_label_report.md`	Counts by label, source, confidence band, season, sample bin, and recording	Decision making before model design
`event_snippet_manifest.csv`	Extractable audio snippets for labelled or weak-labelled events	Species/event layer assets
`activity_label_manifest.csv`	Binary activity/no-activity or event-present labels when species names are missing	Weak activity detector

Decision gates after preprocessing:

Audit result	Recommended Layer C design
Many reliable species labels across enough recordings	Train a small species/event classifier or event likelihood model.

Audit result	Recommended Layer C design
Few species labels but many high-confidence BirdNET events	Use BirdNET pseudo-label retrieval with confidence thresholds.
Mostly score-only events	Use a binary activity layer and avoid species-specific claims.
Labels are highly imbalanced	Use retrieval and rule-based scheduling; avoid supervised multiclass training for MVP.
Event timing is reliable but labels are weak	Use event snippets as generic biological activity textures.

Inputs:

- `season`
- `sample_bin`
- `hour_local`
- `month`
- `day_of_year`
- `temperature_c`
- `humidity_pct`
- `precipitation_mm`
- `days_since_rain`
- audited annotation labels, weak activity labels, or BirdNET pseudo-labels

Recommended MVP implementation:

- First run the annotation audit/preprocessing step.
- Use annotated clips only where labels, timing, and confidence are good enough.
- Add BirdNET pseudo-labels if species coverage is too sparse.
- Train or implement an event likelihood model only after label coverage is known.
- Retrieve real event snippets and place them into the mix with simple scheduling rules.

Comparison:

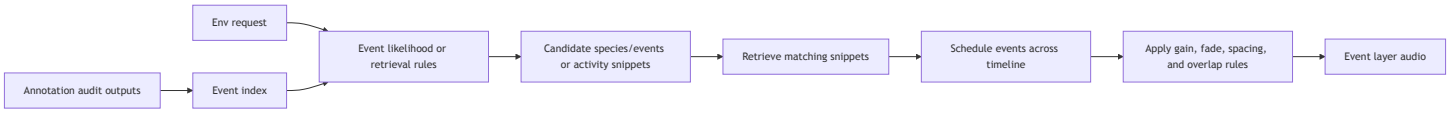
Approach	How it works	Pros	Cons	MVP fit
Annotation audit/index first	Analyse CSV labels, sources, confidence,	Prevents designing the wrong classifier;	Adds a preprocessing step	Required

Approach	How it works	Pros	Cons	MVP fit
	timing, and coverage before training	clarifies what labels are usable		
Annotation retrieval	Select real labelled snippets matching env/time	Realistic and explainable	Existing labels are sparse	Strong if audit confirms enough examples
BirdNET pseudo-label retrieval	Run BirdNET, use high-confidence species labels	Expands label coverage	Pseudo-label errors need thresholding	Strong
Simple probabilistic event model	Estimate event likelihood by season/time/env buckets	Easy to build and explain	Coarse, not deep learning	Strong
MLP classifier on VAE latent	Predict species/event presence from latent z	Uses learned representation	Needs enough labels	Medium
Generative species-call model	Generate calls from scratch	More novel	High risk and data-hungry	Future

MVP recommendation:

```
Use event retrieval and scheduling, not from-scratch species-call generation.
Do annotation preprocessing before deciding whether Layer C is species-specific,
generic biological activity, BirdNET-assisted, or retrieval-only.
```

Example event planning:



Generation Layer D: Mixer and Output Explanation

Purpose:

- Combine layers into one coherent audio file.
- Produce a spectrogram and an explanation so the user understands why the output sounds the way it does.

Mixer responsibilities:

- Match sample rate.
- Trim or loop layers to requested duration.
- Apply fades to avoid clicks.
- Control gain staging to avoid clipping.
- Optionally apply light EQ per layer.
- Export final WAV.
- Generate mel spectrogram preview.
- Return a JSON explanation of selected clips/events.

Example explanation fields:

Field	Meaning
`ambient_source_clips`	Real clips used as the background bed.
`weather_layers`	Wind/rain assets and intensity mapping.
`event_layers`	Species/events selected and why.
`env_match_score`	Similarity between user request and retrieved source clips.
`limitations`	Notes about speculative nature and dataset gaps.

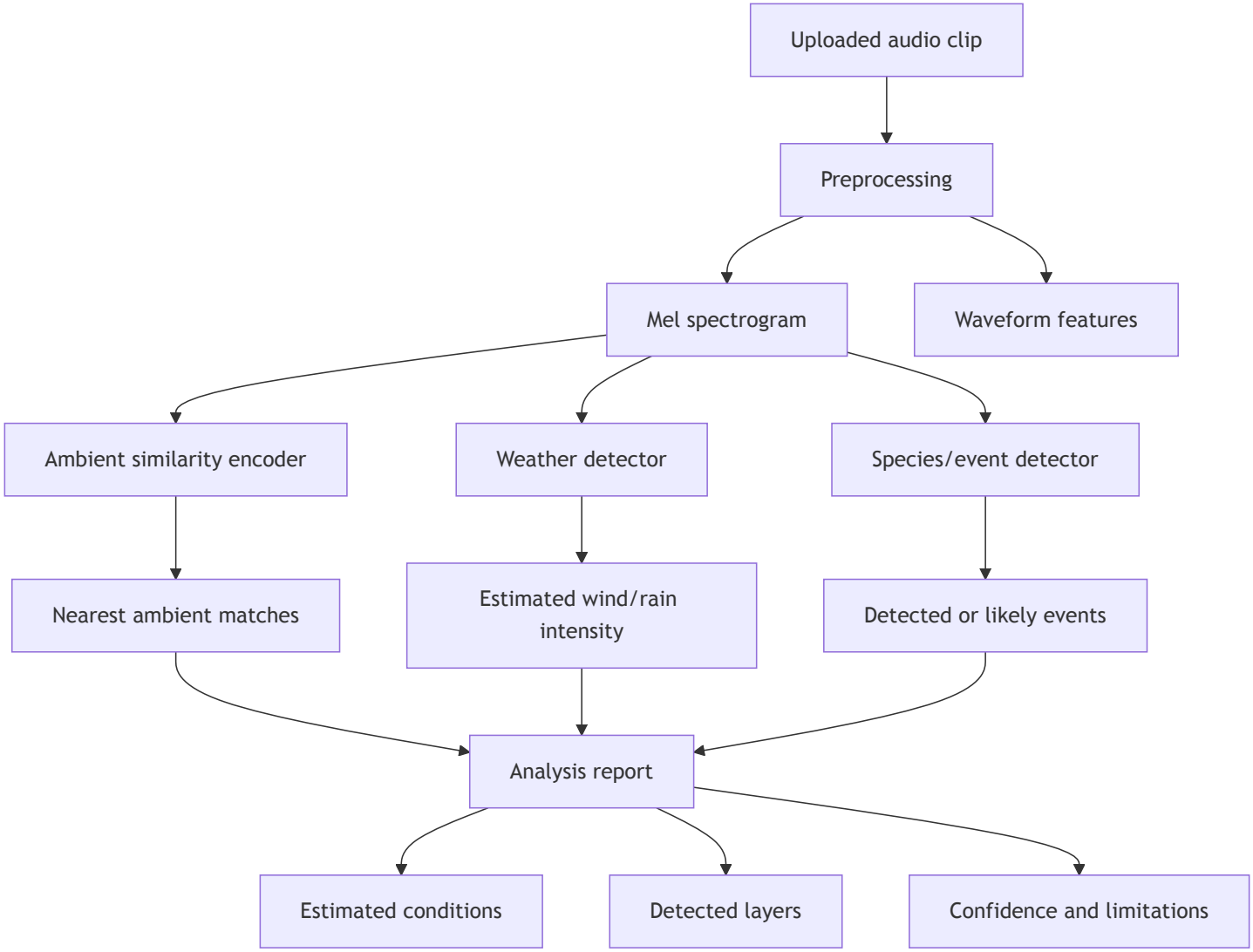
Generation Mode: MVP Build Order

Priority	Component	Reason
1	Ambient retrieval bed	Highest immediate audio realism.
2	Mixer/export pipeline	Needed to combine all later layers.
3	Wind/rain asset layer	Directly maps env controls to audible change.

Priority	Component	Reason
4	Annotation/BirdNET event index	Enables species/event layer.
5	Event planner and scheduler	Makes output ecologically explainable.
6	Optional VAE reconstruction/transformation	Use where it improves variation without hurting quality.
7	Future diffusion/generative modules	Later research extension.

Analysis Mode: Revised Architecture

Analysis mode should not only encode a clip into a latent vector. It should decompose and explain the clip using the same layered logic as generation.



Analysis Component A: Ambient Similarity Encoder

Purpose:

- Locate the uploaded clip in the learned/retrieved soundscape space.
- Estimate broad context: season, diel bin, similar training recordings, and plausible environmental ranges.

Available current implementation:

- `encode_clip()` encodes an uploaded audio clip to latent `mu`.
- `estimate_env_conditions()` compares this latent with `latent_clips.npy`.
- It averages top-k neighbours to estimate environmental conditions.

Comparison:

Approach	How it works	Pros	Cons	MVP fit
Current VAE latent nearest-neighbour	Encode clip, compare latent to training latents	Already implemented, explainable	Accuracy limited by VAE representation	Strong
Audio embedding model such as CLAP/YAMNet	Use pretrained audio embeddings	Strong general audio representation	May not be ecoacoustic/site-specific	Medium
Hand-crafted acoustic indices	Compute ACI, entropy, spectral centroid, band energy	Easy to explain scientifically	Less flexible, not learned	Medium
Custom contrastive encoder	Train retrieval embedding for ecoacoustic similarity	Good long-term	Needs careful training design	Future

MVP recommendation:

```
Keep VAE latent nearest-neighbour for broad context estimation.
Add simple acoustic indices as supporting evidence if time allows.
```

Analysis Component B: Weather Detector

Purpose:

- Detect or estimate audible weather contribution in the uploaded clip.
- Separate direct acoustic weather from indirect environmental context.

Possible outputs:

- `wind_intensity` : none, light, moderate, strong.
- `rain_intensity` : none, light, moderate, heavy.
- confidence score.

Comparison:

Approach	How it works	Pros	Cons	MVP fit
Rule-based spectral heuristics	Use broadband energy, low-frequency modulation, high-frequency rain texture	Fast, no labels needed	Approximate and brittle	Medium
Label curated clips manually	Manually tag wind/rain intensity buckets	Accurate for MVP dataset	Requires small manual effort	Strong
Train classifier on curated labels	CNN/MLP predicts wind/rain class	More scalable	Needs labelled examples	Medium after labels
Use external audio event model	Detect rain/wind from pretrained model	Quick if model works	Domain mismatch possible	Medium

MVP recommendation:

```
Start with small curated labels for wind/rain intensity.
Use these labels both for analysis detection and generation asset retrieval.
```

Analysis Component C: Species and Event Detector

Purpose:

- Identify biologically meaningful events in an uploaded clip.
- Provide a better explanation than "this clip has these weather values."

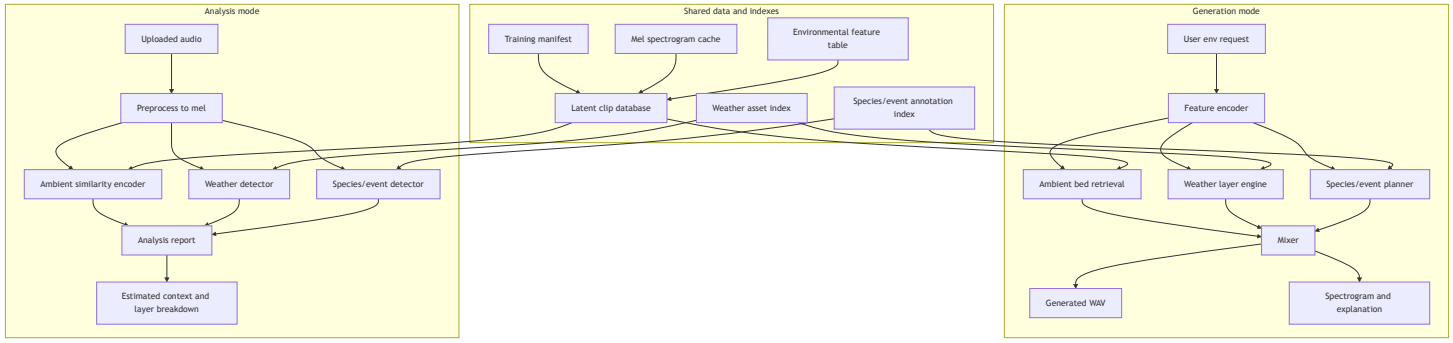
Comparison:

Approach	How it works	Pros	Cons	MVP fit
Existing A2O annotations	Use available annotation CSVs	Grounded in project data	Sparse: only a subset has useful species labels	Medium
BirdNET pseudo-labels	Run BirdNET over clips and uploaded audio	Strong for bird species, expands coverage	Confidence thresholding needed	Strong
Binary activity detector	Treat score-only annotations as activity events	Uses more existing annotation data	Does not identify species	Medium
MLP on frozen VAE latent	Predict event/species vector from latent	Fits A/B/C module plan	Needs enough labels or pseudo-labels	Future/MVP extension

MVP recommendation:

```
Use BirdNET or high-confidence pseudo-labels for species/event analysis.
Use existing annotations as validation or overrides where available.
```

New Unified Architecture Picture



Refined Module Names for MVP

Module	Role	MVP implementation	Future upgrade
Module A: Ambient Representation	Encode/retrieve general site texture	VAE encoder plus nearest-neighbour retrieval	Contrastive ecoacoustic encoder or diffusion prior
Module B: Weather Layer	Add direct rain/wind sounds	Curated weather assets plus parameter mixing	Weather-specific neural generator
Module C: Event Layer	Add species/annotation events	Annotation/BirdNET retrieval plus scheduler	Frozen encoder plus species classifier/generator
Module D: Mixer	Combine and export final soundscape	Librosa/soundfile/pydub-style mixing	Differentiable mixer or DAW-like UI controls
Module E: Analysis Explainer	Explain uploaded/generated audio	Latent nearest-neighbour plus detectors	Full multi-label ecological analysis model

Key Research Framing

The revised MVP should be framed as a layered, environmentally conditioned speculative soundscape system.

This is stronger than claiming the model directly learns a one-to-one mapping from weather to audio.

Suggested wording:

```
Environmental conditions influence soundscape composition through multiple mechanisms.
The MVP models these mechanisms separately: ambient site texture, direct weather acoust
and biologically meaningful event layers. The final soundscape is generated by selectin
transforming, and mixing these layers according to the requested environmental scenario
```

Risks and Mitigations

Risk	Impact	Mitigation
Retrieved audio sounds too close to training data	May appear less generative	Present MVP as retrieval-augmented speculative generation; add transformations and layer mixing.
Species labels are sparse	Weak event layer	Use BirdNET pseudo-labels with confidence thresholds.
Weather labels are missing	Hard to build wind/rain layer	Manually curate a small weather asset set first.
Layer mixing sounds artificial	Reduces demo quality	Use fades, gain normalisation, light EQ, and avoid over-layering.
Env-to-audio claims are too strong	Research criticism	Clearly separate direct acoustic variables from indirect ecological predictors.
VAE output remains poor	Affects generated bed if used directly	Use VAE mainly for embedding/retrieval and optional transformations, not primary audio rendering.

Recommended Immediate Next Steps

Step	Output
1. Build an ambient retrieval function	Given env settings, return top matching real clips and similarity scores.
2. Build a mixer utility	Combine background, weather, and event layers into one WAV.
3. Curate wind/rain asset folders	Small labelled library: no/light/moderate/strong.
4. Add generation API v2	Return final mixed audio plus explanation JSON.
5. Add analysis layer report	For uploaded audio, return ambient match, weather estimate, event estimate, confidence.
6. Add BirdNET or annotation index	Populate species/event layer for generation and analysis.

MVP Success Criteria

Criterion	Target
Audio quality	Output sounds like plausible environmental audio, not electronic machine noise.
Control	Wind/rain/time/season changes produce audible and explainable differences.
Ecological plausibility	Species/events are plausible for the selected season/time/env context.
Explainability	API can report which clips/assets/events were selected and why.
Demo stability	Generation works reliably without depending on fragile latent random sampling.